Analyzing Regional Disparities in Agricultural Wages and Farm Household Income in the Philippines

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1. PROJECT OVERVIEW

This project analyzes agricultural income distribution in the Philippines by examining regional disparities in agricultural wages and farm household income. It aims to identify income imbalances between labor wages and agricultural income, assess whether farming income is sufficient to support household needs, and which regions experience the greatest income inequality and economic vulnerability.

Take note that data gathered is from different years (2000-2016), therefore, findings show historical trends rather than current income conditions and should be interpreted cautiously.

2. DATASET SUMMARY

- Rows: 16
- Columns: 14
- Key Features:
  - **agr_wage_farm_workers_allgender_2015** : Agriculture Wage Rates of Farm Workers (All Gender) in 2015 - UNIT: pesos
  - **agr_wage_farm_workers_male_2015** : Agriculture Wage Rates of Farm Workers (male) in 2015 - UNIT: pesos
  - **agr_wage_farm_workers_female_2015** : Agriculture Wage Rates of Farm Workers (female) in 2015 - UNIT: pesos
  - **avg_annual_total_incm_farm_households_02_03** : Average annual total income for farm households Year 2002-03 - UNIT: pesos
    - Total household income includes the farm income, off-farm income, non-farm income, and other sources of income.
  - **avg_annual_farm_incm_farm_households_02_03** : Average annual total income (farm) for farm households Year 2002-03 - UNIT: pesos
    - Farm income is earned directly from the land or animals like harvesting vegetables and raising livestocks.
  - **avg_annual_off_farm_incm_farm_households_02_03** : Average annual total income (off-farm) for farm households Year 2002-03 - UNIT: pesos

        - Off-farm income is earned from agricultural related activities like selling fertilizers, rice-mills, transporting products outside the farm.
    - **avg_annual_non_farm_incm_farm_households_02_03** : Average annual total income (non-farm) for farm households Year 2002-03 - UNIT: pesos
        - Non-farm income is earned from outside agriculture entirely. For example, having a different job, or running a business.
    - **avg_annual_other_sources_incm_farm_households_02_03** : Average annual total income (other sources) for farm households Year 2002-03 - UNIT: pesos
        - Other sources of income could be earned from external or non-work related activities like remittances, government assistance, & pensions.
    - **avg_rural_income_2000** : Average rural income - UNIT: pesos
    - **total_emply_2016** : Total employment 2016 - UNIT: thousand persons
- Missing Data: 2 values in **avg_rural_income_2000** column

3. EXPLORATORY DATA ANALYSIS USING PYTHON

Data preparation and cleaning was integrated in Python:

- Data Loading: Imported the dataset using pandas.
- Initial Exploration: Used .info() to check data structure and .describe() for summary statistics.
- Data Handling: Checked for null values using is.null().sum() and fill the missing values in **avg_rural_income_2000** using the median of the rural income across regions.
- Feature Engineering:
    - Created **farm_income_share** column by dividing average annual farm income by the average annual total household income.
    - Created **off_farm_income_share** by dividing average annual off-farm income by the average annual total household income.
    - Created **non_farm_income_share** by dividing average annual non-farm income by the average annual total household income.
    - Created **other_sources_income_share** by dividing average annual other sources income by the average annual total household income.
- Database Integration: Connected Python script in jupyter notebook to MySQL and loaded the cleaned DataFrame into the database for SQL analysis

4. DATA ANALYSIS USING SQL

Performed the structured analysis in MySQL to answer key questions:

In the database, a view is created to rename column as follow:

- regDesc **AS region**
- avg_annual_total_incm_farm_households_02_03 **AS total_income**
- avg_annual_farm_incm_farm_households_02_03 **AS farm_income**
- avg_annual_off_farm_incm_farm_households_02_03 **AS off_farm_income**
- avg_annual_non_farm_incm_farm_households_02_03 **AS non_farm_income**
- avg_annual_other_sources_incm_farm_households_02_03 **AS other_sources_income**
- agr_wage_farm_workers_allgender_2015 **AS wage_all**
- gr_wage_farm_workers_male_2015 **AS wage_male**
- agr_wage_farm_workers_female_2015 **AS wage_female**
- avg_rural_income_2000 **AS rural_income,**
- farm_income_share **AS farm_income_share,**
- off_farm_income_share **AS off_farm_income_share,**
- non_farm_income_share **AS non_farm_income_share,**
- other_sources_income_share **AS other_sources_income_share**

KEY QUESTIONS:

1. Which regions have the highest and lowest total household income and which regions have the lowest agricultural wages?

Regions with the highest to lowest farm household income

region	total_income	farm_income	off_farm_income	non_farm_income	other_sources_income
Central Luzon	158130	85048	6073	50413	16596
Cagayan Valley	145809	87353	6909	39894	11653
Calabarzon	143075	64498	8444	48590	21543
Soccsksargen	134406	84331	9347	33658	7070
Car	127067	71142	9731	37178	9016
Mimaropa	126237	74702	6579	35262	9694
Ilocos Region	118637	51544	4358	36054	26681
Davao Region	101673	69496	5593	17573	9011
Zamboanga Peninsula	99327	54135	5946	33431	5815
Caraga	84569	54004	11118	13892	5555
Bicol Region	82988	36579	7348	30381	8680
Eastern Visayas	82503	47359	6066	23812	5266
Northern Mindanao	82299	49988	4256	23279	4776
Western Visayas	79949	41266	6746	21772	10165
Armm	73356	34026	5737	24936	8657
Central Visayas	72177	28074	5574	31055	7474

The 10 lowest agricultural wages across regions

region	wage_all	wage_male	wage_female
Central Visayas	156.17	160.65	150.67
Zamboanga Peninsula	157.37	158.55	152.79
Eastern Visayas	157.49	159.25	112.22
Northern Mindanao	159.12	160.07	151.39
Armm	162.89	163.65	103.81
Soccsksargen	164.77	166.75	159.14
Western Visayas	165.28	167.77	156.96
Bicol Region	167.99	169.95	119.52
Davao Region	168.68	169.83	158.91
Mimaropa	188.73	192.16	170.73

CONCLUSION: Based on the available data from 2002-2003, Central Luzon has the highest total household income among the regions, while Central Visayas ranks the lowest.

Furthermore, Central Visayas having the lowest total household income also records the lowest agricultural wage in the country, based on 2015 data.

2. Are regions where farming is productive (higher farm income) also pay higher wages?

region	farm_income	wage_all	farm_rank	wage_rank
Cagayan Valley	87353	228.77	1	4
Central Luzon	85048	257.97	2	1
Soccsksargen	84331	164.77	3	11
Mimaropa	74702	188.73	4	7
Car	71142	206.68	5	5
Davao Region	69496	168.68	6	8
Calabarzon	64498	230.92	7	3
Zamboanga Peninsula	54135	157.37	8	15
Caraga	54004	194.46	9	6
Ilocos Region	51544	237.26	10	2
Northern Mindanao	49988	159.12	11	13
Eastern Visayas	47359	157.49	12	14
Western Visayas	41266	165.28	13	10
Bicol Region	36579	167.99	14	9
Armm	34026	162.89	15	12
Central Visayas	28074	156.17	16	16

CONCLUSION: Higher farm income does not always equate to higher farm wages. This is evident in regions such as SOCCSKSARGEN and MIMAROPA, where despite ranking high in farm income, suggesting these are rich farming regions, it does not equate to a high income wage.

Therefore, we could say that despite some regions showing strong performance, higher farm income does not correspond to a good paying labor income for farm workers.

3. Which regions have the greatest disparity between farm income and wages?

region	farm_income	annual_wage	wage_ratio
Soccsksargen	84331	60141.05	0.7131547117904449
Davao Region	69496	61568.200000000004	0.8859243697478992
Mimaropa	74702	68886.45	0.9221500093705657
Cagayan Valley	87353	83501.05	0.9559036323881264
Car	71142	75438.2	1.0603890809929437
Zamboanga Peninsula	54135	57440.05	1.0610519996305534
Central Luzon	85048	94159.05	1.1071283275326875
Northern Mindanao	49988	58078.8	1.1618548451628392
Eastern Visayas	47359	57483.850000000006	1.2137893536603392
Calabarzon	64498	84285.79999999999	1.3067971099879063
Caraga	54004	70977.90000000001	1.3143081993926378
Western Visayas	41266	60327.2	1.4619105316725634
Bicol Region	36579	61316.350000000006	1.6762719046447416
Ilocos Region	51544	86599.9	1.6801160173832064
Armm	34026	59454.85	1.74733586081232
Central Visayas	28074	57002.049999999996	2.0304213863361116

Wage ratio = annual_wage / farm_income

CONCLUSION: Regions from SOCCSKSARGEN, Davao Region, Mimaropa, and Cagayan Valley have a low wage-to-farm income ratios, indicating that higher farm income which means there is a high farming productivity does not necessarily translate into higher wages for farm workers. Meanwhile, regions such as Central Visayas show a high wage-to-farm ratio, but it does not mean that they have the highest agricultural wage, in fact they are the lowest. Instead this shows that the region has a relatively low farm income, suggesting that the region has a weak farm productivity compared to other regions.

4. Are there wage differences between male and female workers across regions, And if so, which regions have the largest gap?

region	wage_male	wage_female	wage_gap	wage_gap_amount
Armm	163.65	103.81	Male higher	59.84
Calabarzon	231.45	172.04	Male higher	59.41
Central Luzon	259.04	204.73	Male higher	54.31
Bicol Region	169.95	119.52	Male higher	50.43
Eastern Visayas	159.25	112.22	Male higher	47.03
Ilocos Region	239.19	199.04	Male higher	40.15
Cagayan Valley	232.64	199.98	Male higher	32.66
Mimaropa	192.16	170.73	Male higher	21.43
Car	211.04	195.62	Male higher	15.42
Davao Region	169.83	158.91	Male higher	10.92
Western Visayas	167.77	156.96	Male higher	10.81
Central Visayas	160.65	150.67	Male higher	9.98
Northern Mind...	160.07	151.39	Male higher	8.68
Caraga	195.44	187.78	Male higher	7.66
Soccsksargen	166.75	159.14	Male higher	7.61
Zamboanga Pe...	158.55	152.79	Male higher	5.76

CONCLUSION: There is an observable wage difference between the male and female workers, showing a huge gap in wages ranging from as high as ₱59.84 php to as low as ₱5.76. It is shown that the ARMM, CALABARZON, and Central Luzon regions show the biggest gender wage gap, respectively.

Overall, male agricultural workers are consistently paid more than the female workers highlighting that there is a disparity between gender in the sector.

5. How much of total household income comes from farming alone, and is it sufficient?

Percentage of household income from farming alone

region	total_income	farm_income_share
Armm	73356	0.46
Bicol Region	82988	0.44
Cagayan Valley	145809	0.6
Calabarzon	143075	0.45
Car	127067	0.56
Caraga	84569	0.64
Central Luzon	158130	0.54
Central Visayas	72177	0.39
Davao Region	101673	0.68
Eastern Visayas	82503	0.57
Ilocos Region	118637	0.43
Mimaropa	126237	0.59
Northern Mind...	82299	0.61
Soccsksargen	134406	0.63
Western Visayas	79949	0.52
Zamboanga Pe...	99327	0.55

Is farm income sufficient?

region	farm_income	rural_income	farm_income_sufficiency
Cagayan Valley	87353	94212	Farm income insufficient
Central Luzon	85048	125626	Farm income insufficient
Soccsksargen	84331	73428	Farm income sufficient
Mimaropa	74702	73819.5	Farm income sufficient
Car	71142	95635	Farm income insufficient
Davao Region	69496	79983	Farm income insufficient
Calabarzon	64498	73819.5	Farm income insufficient
Zamboanga Peninsula	54135	64413	Farm income insufficient
Caraga	54004	63574	Farm income insufficient
Ilocos Region	51544	103915	Farm income insufficient
Northern Mindanao	49988	73966	Farm income insufficient
Eastern Visayas	47359	68077	Farm income insufficient
Western Visayas	41266	76488	Farm income insufficient
Bicol Region	36579	72626	Farm income insufficient
Armm	34026	73673	Farm income insufficient
Central Visayas	28074	68447	Farm income insufficient

CONCLUSION: Income from farming makes up the largest share of total farm household income, with an average income share of 54%, and some regions reaching as high as 68%. However, in most regions, income obtained alone from farming is insufficient when compared to the average rural income, indicating that households rely on additional income sources beyond farming. Only the regions of SOCCSKSARGEN and MIMAROPA show sufficient annual farm income (₱84,331 and ₱74,702 respectively), indicating that their farming income is adequate relative to rural income standards.

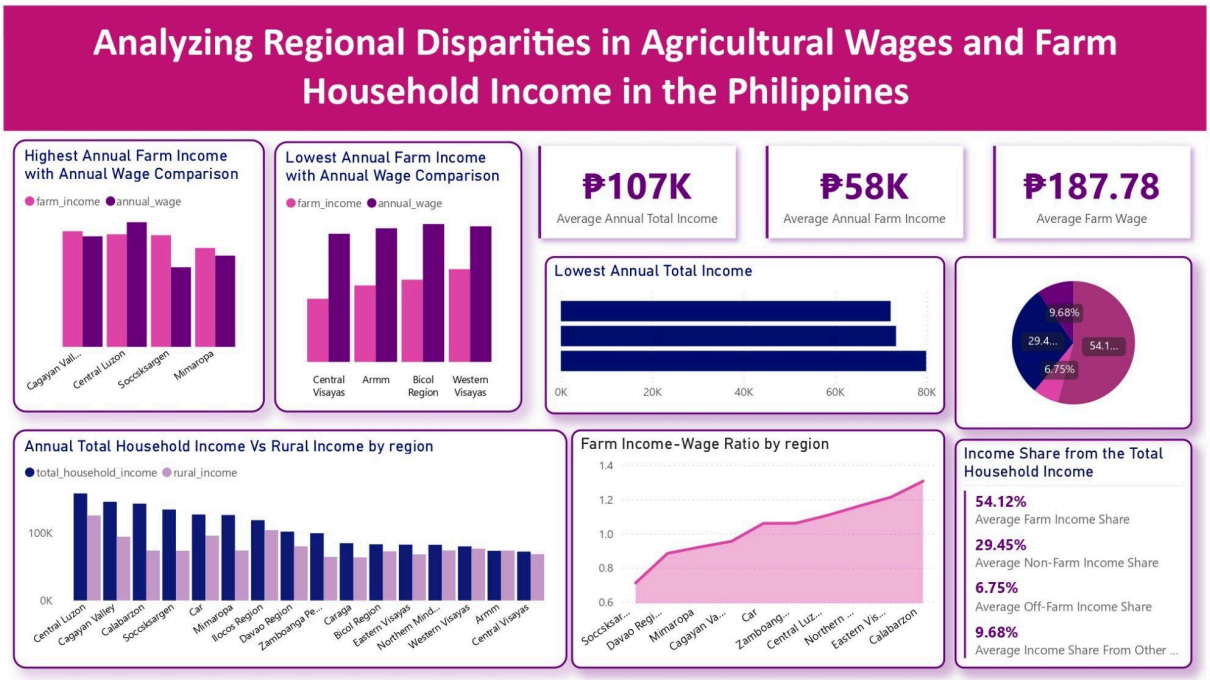
Take note that the data used are from a different year period and might not reflect the current economic condition. Additionally, rural income is used as a reference point and not an actual standard for livable household income.

6. Comparing the total household income with average rural income across regions, which regions are above & below the average rural income?

region	income_status
Armm	Below
Bicol Region	Above
Cagayan Valley	Above
Calabarzon	Above
Car	Above
Caraga	Above
Central Luzon	Above
Central Visayas	Above
Davao Region	Above
Eastern Visayas	Above
Ilocos Region	Above
Mimaropa	Above
Northern Mind...	Above
Soccsksargen	Above
Western Visayas	Above
Zamboanga Pe...	Above

CONCLUSION: Although farm income alone is generally not sufficient relative to rural income standards, the total household income, comprising farm income, non-farm income, off-farm income, and other sources shows that most regions are above the average rural income standards of their region. Only the ARMM region falls below the average rural income.

5. DASHBOARD IN POWERBI



6. INSIGHTS AND SUGGESTIONS

- **Mismatch between productivity and wages:**
  High farm income regions do not always provide higher wages to workers
- **Income inequality exists across regions:**
  Some regions show strong agricultural output but lower worker compensation
- **Dependence on non-farm income:**
  Many households rely on additional income sources beyond farming
- **Farm income insufficiency:**
  In most regions, farming alone is not enough to meet rural income standards

SUGGESTIONS:

- **INCOME SUPPORT:** We should promote policies, as well as the government should create and improve these policies that support access to higher farm income, focusing on lowering production costs through technical assistance and provide better pricing mechanisms.
- **ACCESSIBLE EQUIPMENT, MARKET, & AGRICULTURAL SERVICES :** The government should focus on expanding programs that provide free and accessible modern equipment, as well as, improved access to markets and

services. Farmers should have easy access to credit in government banks with low-interest rates to help in the procurement of farming inputs and tools. The Department of Agriculture should look for ways that could improve life and improve farmer productivity.

- **GENDER WAGE EQUALITY:** There is an apparent gender wage disparity in the agricultural sector, favoring higher income for the men than the women. To address this issue, there should be a need to strengthen and enforce policies that ensure fair and equal compensation regardless of gender. Also, we should enforce programs that promote women participating in higher paying agricultural roles.